Change title

AI Coding Agents

A reductive account for physicists

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Add affiliation, LUH + logo
innovailia

WHY THIS TALK?

Put here the GPT 5.2

Physics Result press release

Three moments

Nov 2022
ChatGPT launches

Q1 2025
Claude Code, Cursor, Copilot

Q4 2025
Models cross a threshold

Eliminate data

Quick survey : Experience

~~Raise your hand if you have…~~

move up

- ~~used~~ ChatGPT or Claude , Gemini or Grok?
- ~~used it~~ to write code IDE + LLM
- ~~used~~ Claude Code, Cursor, or Copilot what agent?

If Experimented with

chat in Q1 2025

dismissed LLMs!

I did.

I was a skeptic.

I am still annoyed at models of Q1 2025-

"LLMs are incredibly capable but maximally untrustworthy."

→ They lie, cheat, steal, flatter

→ All terrible for hard repro science

The evidence

Seductively correct equations that are nonsense.

um

Push back on a valid proof ~~and it~~ finds a “bug” that isn’t there.

I got burnt. Then I got curious.

Good mental model of

LLMs:

slightly deranged hyper motivated

MSc student

Picture?

LLMs are useful.

But

1000+ hours of deliberate practice.

Nov 2025 – today.

deliberate practice req'd

1. ✓ 2. pay $ 3. Need to install codex agent; full access

Correct workflow → amazing things are possible

What I built

Pray

vibefeld	Go, 112k lines	adversarial proof verification
Lyr.jl	Julia	volume renderer, GR ray tracing
af-tests	Lean 4, 0 sorries	formal proofs in operator algebras
convexfeld	C, 37k lines	LP solver from scratch
Nano MIPS	Verilog	pipelined CPU on $10 FPGA

Showcase later.

Vibefeld

Formalisation is real (Lean)

What I built *(continued)*

write new science

PicoGPT.jl	Julia	GPT with manual backprop (1 day)
vectorfeld	TypeScript + Rust	vector graphics editor (1 day)
datapipeline	Python + DuckDB	3.7M research grants (5 days)
alethfeld	Clojure, 134 ★	adversarial proofs (open source)

All with basic tooling
No extra subscriptions.

Accelerating research is possible

All of this was built with AI coding agents.

~~ERA~~ of Automated research is possible. With correct tools AND expert guidance (Human-gated)

~~Let me show you the entire trick.~~

I hate magic & mystery

$\Longrightarrow$ LLMs can be understood.

Ground up $\rightarrow$ Coding agents

WHAT IS AN LLM?

~~The observable interface~~

From outside looking in:

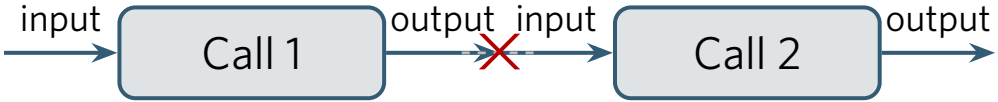
Large Language model

```
f : String -> String
```

From outside the API boundary, this is the entire observable behaviour.

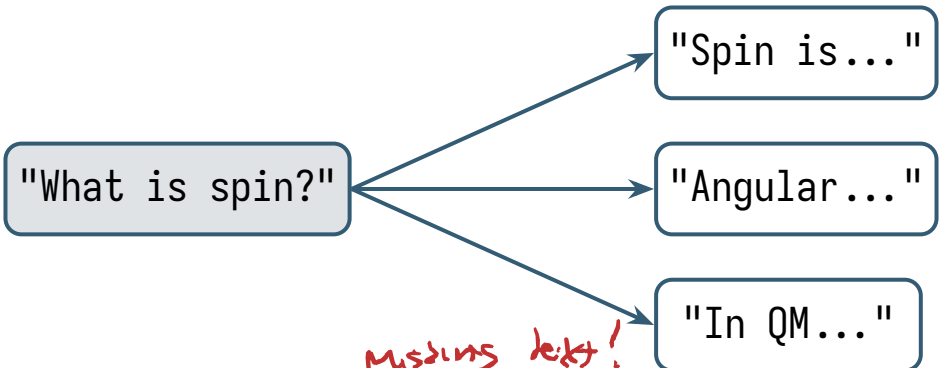
No hidden state is carried between calls.

Stateless



Each evaluation is independent. No memory between calls.

Nondeterministic



Missing text!

Same input, different outputs. The model samples from $P(\text{next token} \mid \text{context})$.

Temperature

slide in between

LLM → prob distr. → samples from it

then samples →

$$P(\text{token}_i) = \frac{\exp(\ell_i/T)}{\sum_j \exp(\ell_j/T)}$$

Direct analogy to the Boltzmann distribution.

Temperature limits

$T \to 0$: deterministic (ground state)

$T \to \infty$: uniform (infinite temperature)

Actually, not strings → strings But rather [Token]^×n

Tokens

Not characters. Not words.

Subword chunks from a fixed vocabulary V of $\sim$100k entries.

```
'theoretical physics'  ->  ['theor', 'etical', ' physics']
```

LLM =

One forward pass through a frozen neural network.

No learning. No state mutation.

Summary

An LLM is a stateless, nondeterministic function

$$f : \text{String} \rightarrow \text{String}$$

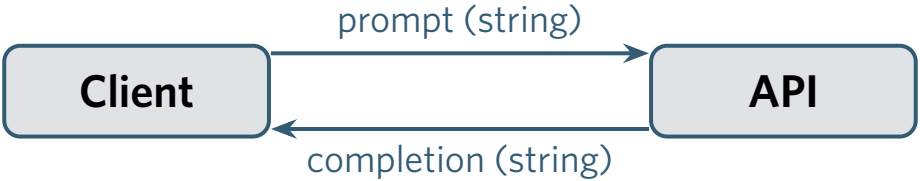
that samples from $\mathrm{P}(\text{next_token} \mid \text{context})$

via a single forward pass through a frozen neural network.

THE ILLUSION OF CHAT

You interact with LLM via

A single API call :



One string in. One string out. That is the entire interaction.

The HTTP reality

```
POST /v1/messages
{
  "model": "claude-sonnet-4-6-20250627",
  "messages": [
    {"role": "user",
     "content": "What is a quark?"}
  ]
}
```

How chat works

```
Turn 1:  [user_1]
Turn 2:  [user_1, asst_1, user_2]
Turn 3:  [user_1, asst_1, user_2, asst_2, user_3]
```

The client re-sends the **entire** history every time.

The LLM has no memory.

The client maintains the illusion.

The context window

The input has a maximum length in tokens.

When the conversation exceeds it, something must be dropped.

Typical sizes: 8k, 128k, 200k tokens.

free chat

Claude

Frontier model

Another Slide!

Context rot / Amnesia

The system prompt

A hidden preamble prepended to every call.

It shapes behaviour, persona, and constraints.

It is just more text.

→ link to Anthropic leaked prompt

web page address

Chat is a loop.

Memory is an illusion maintained by the client.

FROM FUNCTION TO AGENT

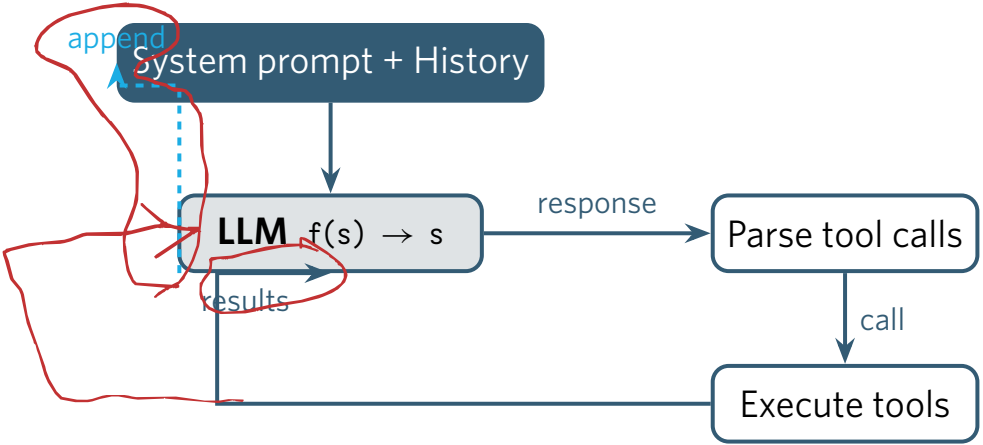
Extra slide: Web interface Demos chat
Terminal open for API call

The agent loop

```
while not done:
    response = llm(system_prompt + history)
    tool_calls = parse(response)
    results = execute(tool_calls)
    history += [response, results]
```

This is the complete architecture of every coding agent.

The agent loop



~~Tool definition~~ Tools

You describe tools in the system prompt.

The LLM generates structured text requesting a tool call.

Your scaffolding parses and executes it.

The LLM never executes anything.

Two tools

```
read_file(path)            -> contents
write_file(path, content) -> ok
```

That is enough to do useful work.

Claude Code, Cursor, Copilot:

the same loop, more tools, better prompts.

LIVE DEMO

Watch the agent work

WHAT SHOULD WE TRY?

Option 1: LaTeX lint

Parse a `.tex` file.

Find every undefined `\ref` and `\cite`.

Report what is broken.

Option 2: Data to figure

Read a CSV. Fit a power law.

Generate a publication-quality plot with error bars.

Save as PDF.

Option 3: DOI to BibTeX

Given a list of DOIs, fetch metadata.

Produce a consistently formatted `.bib` file.

Option 4: Seminar to calendar

Scrape a seminar schedule from a URL.

Produce an `.ics` calendar file.

Option 5: Batch rename

I describe a naming convention in English.

The agent renames 200 files to match.

Option 6: Dimensional analysis

Read a paper draft.

Check that every equation is dimensionally consistent.

~~Raise your hand.~~